4.2 Forces: Density & Pressure

Question Paper

Course	CIE A Level Physics
Section	4. Forces, Density & Pressure
Topic	4.2 Forces: Density & Pressure
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The density of water is 1.0 g cm^{-3} and the density of glycerine is 1.3 g cm^{-3}

Water is added to a measuring cylinder containing 40 cm^3 of glycerine so that the density of the mixture is 1.1 g cm^{-3}. Assume that the mixing process does not change the total volume of the liquid

What is the volume of water added ?

A. 40 cm^3

B. 44 cm^3

C. 52 cm^3

D. 80 cm^3

[1 mark]

Question 2

A cylindrical block of wood has cross-sectional area *A* and weight *W*. It is totally immersed in water with its axis vertical. The block experiences pressures p_t and p_b at its top and bottom surfaces respectively.

Which expression is equal to the upthrust on the block ?

A. $(p_b - p_t)A + W$

B. $(p_b - p_t)$

C. $(p_b - p_t)A$

D. $(p_b - p_t)A - W$

[1 mark]

Question 3

Full-fat milk is made up of fat-free milk mixed with fat.

A volume of 1.000×10^{-3} m^3 of full-fat milk has a mass of 1.035 kg. It contains 4.00% fat by volume.

The density of fat-free milk is 1.040×10^3 kg m^{-3}

What is the density of fat ?

A. 1.25×10^2 kg m^{-3}

B. 9.15×10^2 kg m^{-3}

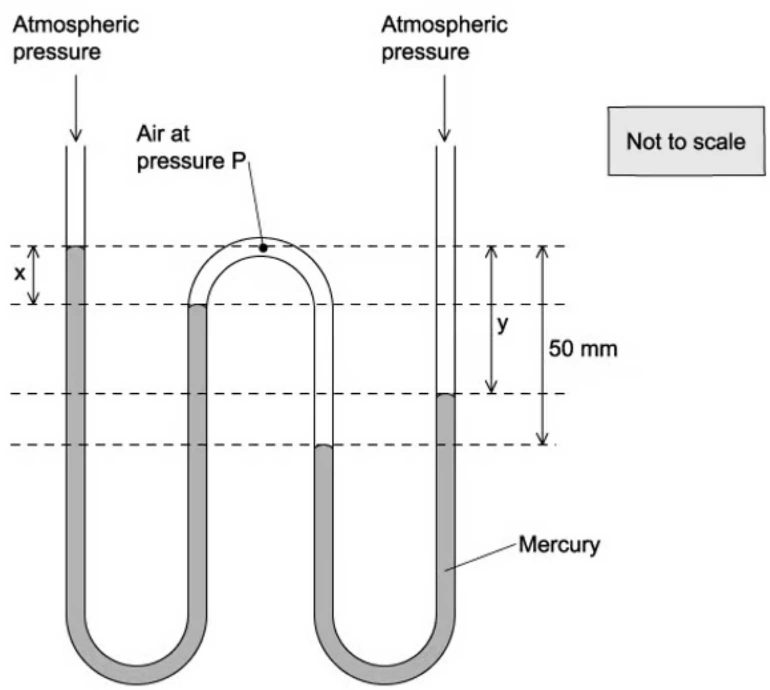
C. 9.28×10^2 kg m^{-3}

D. 1.16×10^3 kg m^{-3}

[1 mark]

Question 4

A W-shaped tube contains two amounts of mercury, each open to the atmosphere. Air at pressure *P* is trapped in between them. The diagram shows two vertical distances *x* and *y*.



Atmospheric pressure is equal to the pressure that would be exerted by a column of mercury of height 760 mm. The pressure *P* is expressed in this way.

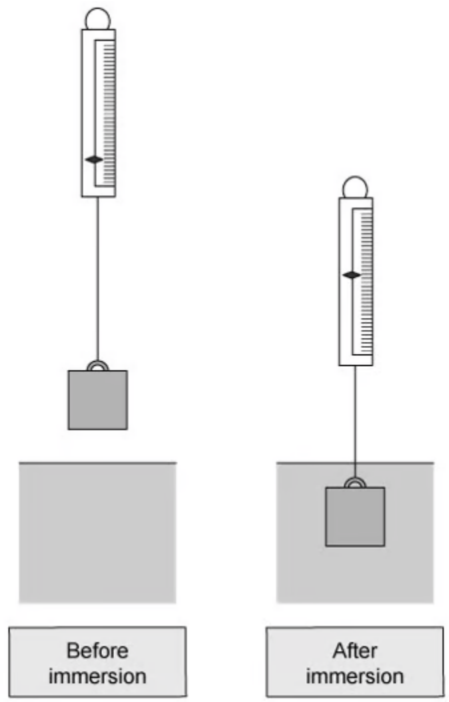
Which values of *x*, *y* and *P* are possible ?

	***x* / mm**	***y* / mm**	***P* / mm of Mercury**
A	20	20	780
B	20	30	780
C	30	20	810
D	30	30	790

[1 mark]

Question 5

The diagram shows a metal cube of density ρ suspended from a spring balance before and during immersion in water. Each face of the cube has a surface area A and each edge has length h.



A reduction in the balance reading occurs as a consequence of the immersion. The cube experiences pressures p_1 and p_2 at its top and bottom surfaces respectively.

What is the value of the upthrust on the metal cube during immersion ?

A. $Ah\rho g$

B. $Ah\rho g + p_1 A$

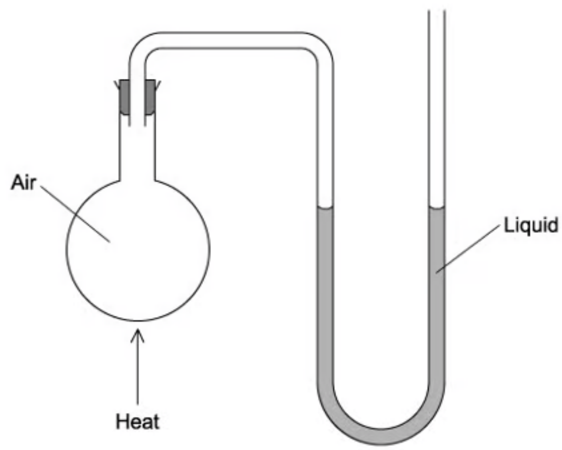
C. $p_2 A$

D. $p_2 A - p_1 A$

[1 mark]

Question 6

The diagram shows a flask connected to a U-tube containing liquid. The flask contains air at atmospheric pressure.



The flask is now gently heated and the liquid level in the right-hand side of the U-tube rises through a distance *h*. The density of the liquid is ρ.

What is the increase in pressure of the heated air in the flask?

A. $h\rho$

B. $\frac{1}{2}h\rho g$

C. $h\rho g$

D. $2h\rho g$

[1 mark]

Question 7

Two solid substances P and Q have atoms of mass M_P and M_Q respectively. They have n_P and n_Q atoms per unit volume.

The density of P is greater than the density of Q.

Which expression **must** be correct?

A. $M_P > M_Q$

B. $n_P > n_Q$

C. $M_{:P} n_P > M_Q n_Q$

D. $\frac{M_P}{n_P} > \frac{M_Q}{n_Q}$

[1 mark]

Question 8

Liquid Q has twice the density of liquid R.

At depth *x* in liquid R, the pressure due to the liquid is 4 kPa.

At what depth in liquid Q is the pressure due to the liquid 7 kPa?

A. $\frac{2x}{7}$

B. $\frac{7x}{8}$

C. $\frac{8x}{7}$

D. $\frac{7x}{2}$

[1 mark]

Question 9

A child drinks a liquid of density ρ through a vertical straw.

Atmospheric pressure is p_0 and the child is capable of lowering the pressure at the top of the straw by 10%. The acceleration of free fall is g.

What is the maximum length of straw that would enable the child to drink the liquid?

A. $\dfrac{p_0}{10\rho g}$

B. $\dfrac{9p_0}{10\rho g}$

C. $\dfrac{p_0}{\rho g}$

D. $\dfrac{10p_0}{\rho g}$

[1 mark]

Question 10

Icebergs typically float with a large volume of ice beneath the water. Ice has a density of 917 kg m^{-3}.

The density of seawater is 1020 kg m^{-3}

What fraction of the ice is submerged underwater?

A. $0.05\ V_i$

B. $0.10\ V_i$

C. $0.90\ V_i$

D. $0.95\ V_i$

[1 mark]

Model Answers: Hard

1

The correct answer is **D** because:

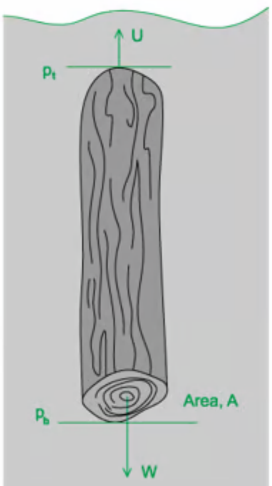
- When the water and the glycerin are added together, the density of the mixture is obtained using $\frac{total\ mass}{total\ volume}$
- For water:
  - Density of water, $\rho_w = 1.0\ g\ cm^{-3}$
  - Mass of water $= m_w$
  - Volume of water $= V_w$
- For glycerine:
  - Density of glycerine, $\rho_g = 1.3\ g\ cm^{-3}$
  - Mass of glycerine $= m_g$
  - Volume of glycerine, $V_g = 40\ cm^3$
- The mass of the glycerine can be found using
  - Mass, $m_g = \rho_g \times V_g = 1.3 \times 40 = 52\ g$
- Consider the mixture:
  - Density of mixture, $\rho = 1.1\ g\ cm^{-3}$
  - Total mass of mixture $= m_w + m_g = m_w + 52$
- The mixing process does not change the total volume of the liquid
  - Total volume of mixture $= V_w + 40$
- Density of mixture $= \frac{total\ mass}{total\ volume}$
  - $1.1 = \frac{(m_w + 52)}{(V_w + 40)}$
  - $V_w + 40 = \frac{(m_w + 52)}{1.1}$ **(1)**
- Consider the density of water
  - $m_w = \rho_w \times V_w = 1.0\ V_w = V_w$ **(2)**

- Replace m_w by V_w in equation **(1)**
  - $V_w + 40 = \frac{(V_w + 52)}{1.1}$
  - $V_w = \frac{V_w}{1.1} + \frac{52}{1.1} - 40$
  - $V_w - \frac{V_w}{1.1} = \frac{52}{1.1} - 40$
  - $V_w\left(1 - \frac{1}{1.1}\right) = \frac{80}{11}$
  - $V_w = 80\ cm^3$

2

The correct answer is **C** because:

- The upthrust of the block is the net vertical force acting on the block
- Pressure is equal to $p = \frac{F}{A}$
- Therefore, force is equal to $F = p \times A$ (where A is the cross-sectional area)
- Now, because the bottom of the block is deeper into the water than the top of the block, $p_b > p_t$
- So, the force due to the pressure at the bottom of the block $= p_b \times A$
- And the force due to the pressure at the top of the block $= p_t \times A$
- Upthrust on block $= p_bA - p_tA = (p_b - p_t)\,A$



A & D are incorrect because	these expressions correctly (D) and incorrectly (A) descr on the wooden cylinder
B is incorrect because	this expression describes the difference in pressure bet bottom of the wooden cylinder

3

The correct answer is **B** because:

- Density of full-fat milk $= \dfrac{total\ mass}{total\ volume}$
- Total mass = mass of fat + mass of fat-free milk
- Total volume = volume of fat + volume of fat-free milk
- Consider the full-fat milk
  - Mass of full-fat milk $= 1.035\ kg$
  - Volume of full-fat milk $= 1.000 \times 10^{-3}\ m^3$
  - Density of full-fat milk $= \dfrac{mass}{volume} = \dfrac{1.035}{1.000 \times 10^{-3}} = 1035\ kg\ m^{-3}$
- The full-fat milk contains 4.00% fat by volume
  - Volume of fat $= 0.04 \times 1.000 \times 10^{-3} = 4.0 \times 10^{-5}\ m^3$
- Therefore, the fat-free milk makes up the remaining 96% of the volume
  - Volume of fat-free milk $= 0.96 \times 1.000 \times 10^{-3} = 9.6 \times 10^{-4}\ m^3$
- Given that the density of fat-free milk $= 1.040 \times 10^3\ kg\ m^{-3}$
  - Mass of fat-free milk $= density \times volume = (1.040 \times 10^3) \times (9.6 \times 10^{-4}) = 0.9984\ kg$
- Total mass = mass of fat + mass of fat-free milk
- Mass of fat $= 1.035 - 0.9984 = 0.0366\ kg$
  - Density of fat $= \dfrac{mass}{volume} = \dfrac{0.0366}{4.0 \times 10^{-5}} = 915\ kg\ m^{-3}$
- The density of fat is therefore equal to 9.15×10^2 kg m^{-3}

Exam Tip: the question indicates that up to four significant figures should be considered as quantities such as volume are given as 1.000×10^{-3} rather than 1.0×10^{-3}, not doing so could affect the final answer you obtain

4

The correct answer is **B** because:

- Atmospheric pressure P_A = column of mercury of height 760mm
- Compare the left part with the middle (left) part
- Since the position of the mercury column is lower in the middle, the pressure P is greater than the atmospheric pressure acting at the left 'open' tube
  - $P - P_A = x$
  - $P - 760 = x$ **(1)**
- Now, compare the right part of the diagram with the middle (right) part
  - $P - P_A = 50 - y$
  - $P - 760 = 50 - y$
  - $P + y = 810$ **(2)**
- The values from the table should satisfy both equation **(1)** and **(2)**
- Consider option B:
  - $x = 20$, $P = 780$
- Equation **(1)**: $780 - 760 = 20$
  - $y = 30$, $P = 780$
- Equation **(2)**: $780 + 30 = 810$
- These values satisfy equations **(1)** and **(2)** so option B is correct

A is incorrect because	$x = 20$, $P = 780$ Equation **(1)**: $780 - 760 = 20$ ✓ $y = 20$, $P = 780$ Equation **(2)**: $780 + 20 = 810$ ×
C is incorrect because	$x = 30$, $P = 810$ Equation **(1)**: $810 - 760 = 30$ × $y = 20$, $P = 810$ Equation **(2)**: $810 + 20 = 810$ ×
D is incorrect because	$x = 30$, $P = 790$ Equation **(1)**: $790 - 760 = 30$ ✓ $y = 30$, $P = 790$ Equation **(2)**: $790 + 30 = 810$ ×

5

The correct answer is **D** because:

- The upthrust of the block is the net vertical force acting on the block
- Pressure is equal to $p = \frac{F}{A}$

- Therefore, force is equal to $F = p \times A$ (where A is the cross-sectional area)
- Now, because the bottom of the block is deeper into the water than the top of the block, $p_2 > p_1$

the block, $p_2 > p_1$

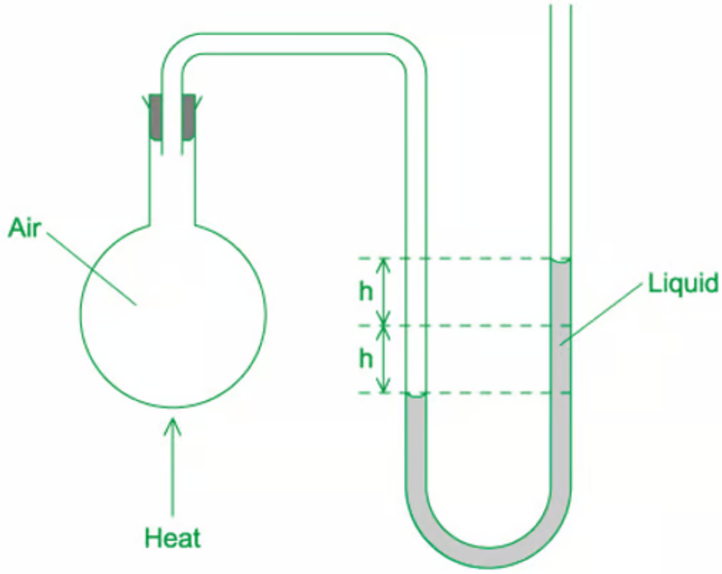
- So, the force due to the pressure at the bottom of the block $= p_2 \times A$
- And the force due to the pressure at the top of the block $= P_1 \times A$
- Upthrust on block $= p_2 A - p_1 A$

A & B are incorrect because	the term $Ah\rho g$ describes the force on the cube at a ce water, the value of this changes depending on the dept upthrust stays the same, no matter the depth
C is incorrect because	the term $p_2 A$ describes the force on the bottom surfa upthrust as, again, the value of this changes depending the pressure change

6

The correct answer is **D** because:

- The equation of hydrostatic pressure is $\Delta P = (\Delta h)\rho g$
- The heating causes the liquid in the right-hand side to rise a distance *h* from its original position
- But this also causes the liquid in the left-hand side to sink by a distance *h* from its original position
- So, the difference in height of the liquid is $\Delta h = h + h = 2h$
- Therefore, the increase in pressure is
  - $\Delta P = (\Delta h)\rho g = 2h\rho g$



7

The correct answer is **C** because:

- The mass of one atom of P and Q are M_P and M_Q respectively
- The number of atoms per unit volume of P and Q are n_P and n_Q respectively
- Total mass per unit volume $= Mn$
- Density $= \frac{total\ mass}{volume}$
    - Density of P $= M_P n_P$
    - Density of Q $= M_Q n_Q$
- The density of P is greater than the density of Q so
    - $M_P n_P > M_Q n_Q$

A & B are incorrect because	it cannot be said conclusively whether or not $M_P > M_Q$ because all we know is that the density of P is greater tha we cannot conclude the individual values of M and n fror alone
D is incorrect because	n represents the number of atoms per unit volume so $\frac{M}{n}$ $mass \times volume$, which is an incorrect expression for der

8

The correct answer is **B** because:

- The equation of hydrostatic pressure is $P = h\rho g$
- For liquid R:
  - Pressure, $P_R = h\rho_R g = x\rho_R g = 4$
  - Density, $\rho_R = \dfrac{4}{xg}$ **(1)**
- For liquid Q:
  - Density, $\rho_Q = 2\rho_R$
  - Pressure, $P_Q = h\rho_Q g = h(2\rho_R)g = 7$
- So, depth in liquid Q:
  - $h = \dfrac{7}{2\rho_R g}$ **(2)**
- Substituting equation **(1)** into **(2)**:
  - $h = \dfrac{7}{2g\left(\dfrac{4}{xg}\right)} = \dfrac{7}{2}\left(\dfrac{4}{x}\right)^{-1} = \dfrac{7x}{8}$

Exam Tip: in questions such as these, identify the link between the two quantities in order to equate them. For example, in this question, ρ is the quantity that links R and Q, so identifying ρ in terms of the other known variables is a crucial step

9

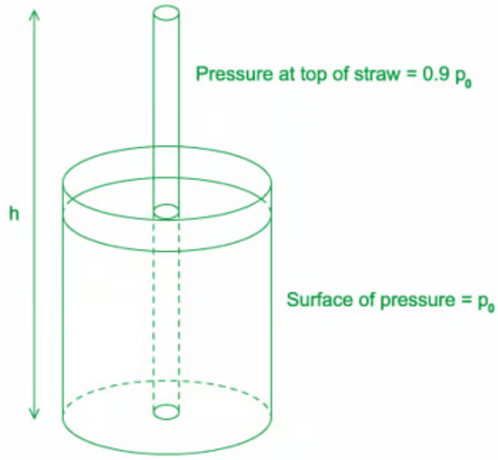
The correct answer is **A** because:

- The equation of hydrostatic pressure is $P = h\rho g$
- Imagine a beaker containing the liquid and a vertical straw (not inclined)
- Atmospheric pressure p_0 acts on the surface of the liquid
- The child is capable of lowering the pressure at the top of the straw by 10%
- So, the child is drinking from the top of the straw with a pressure of $(1 - 0.1)p_0 = 0.9p_0$
- As the child drinks, the liquid rises up the straw of length h
- The difference in pressure at the surface if the liquid in the beaker p_0 and the

pressure at the top of the straw $0.9p_0$ is equal to the pressure due to the column of liquid in the straw (from the equation of hydrostatic pressure)

- $p_0 - 0.9p_0 = h\rho g$
- $h\rho g = 0.1p_0 = \dfrac{p_0}{10}$
- $h = \dfrac{p_0}{10\rho g}$

- The length of the straw is equal to $h = \dfrac{p_0}{10\rho g}$



10

The correct answer is **C** because:

- According to Archimedes' principle, the buoyant force on an object equals the weight of the fluid it displaces
- The iceberg has weight $W_i = M_i g$ and the buoyant force is equal to the weight of the displaced water, $W_w = M_w g$
- Furthermore, since the iceberg is floating, its weight exactly balances the buoyant force:
  - $W_w = W_i$
  - $M_w g = M_i g$

- Density is equal to $\frac{mass}{volume}$ so mass = density × volume
  - $V \quad \rho \quad g = V \, . \, \rho \, . \, g$

10

The correct answer is **C** because:

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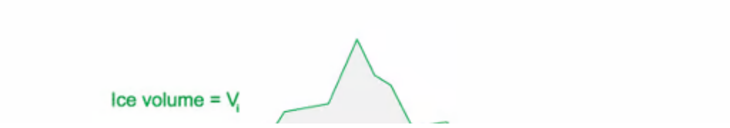
- The iceberg has weight $W_i = M_i g$ and the buoyant force is equal to the weight of the displaced water, $W_w = M_w g$
- Furthermore, since the iceberg is floating, its weight exactly balances the buoyant force:
    - $W_w = W_i$
    - $M_w g = M_i g$
- Density is equal to $\frac{mass}{volume}$ so mass = density × volume
    - $V_w \rho_w g = V_i \rho_i g$
    - $V_w = \frac{\rho_i}{\rho_w} V_i$
- So, the fraction of ice underwater $\frac{V_w}{v_i}$ is given by the ratio of densities
    - $\frac{V_w}{V_i} = \frac{\rho_i}{\rho_w} = \frac{917}{1020} = 0.899$
    - $V_w = 0.90\ V_i$

Therefore, 90% of an iceberg's volume (and mass) is submerged underwater


Ice volume = V_i